General instructions:

The goal of this project is for you to handle working on a problem in deep learning on your own, including downloading data, creating environments, suggesting solutions, implementing them, and fixing their issues. We are not expecting results worthy of publication as novel work, but rather actual experimentation on your part, showing creative thought and evident effort.

In the next sections, we will provide you with three ideas for problems to tackle, which we recommend you try to solve. You can also choose to tackle a different problem - notes on that later.

You are also welcome to use published papers, published code, or AI to help with your work, but you must disclose all such uses in your report.

Your submission must include a single zip file named <id1>-<id2>.zip, where id1 and id2 are the IDs of you and your partner. This folder must include the following files:

- report.pdf – see below
- README.md – a file explaining how to run your code.
- not_our_work.txt – a plain text file listing any files you wish to be excluded from automatic checks for internet or AI use (see last section).
- code/ – a folder containing all your code files.

The submission follows the general course late submission policy and should be submitted via the grades website, with the deadline being 5.3.2026 at 23:59. Late submissions will remain open nearly indefinitely for reserve duty students, but note that you will receive a failing grade if you submit more than a week after the deadline.

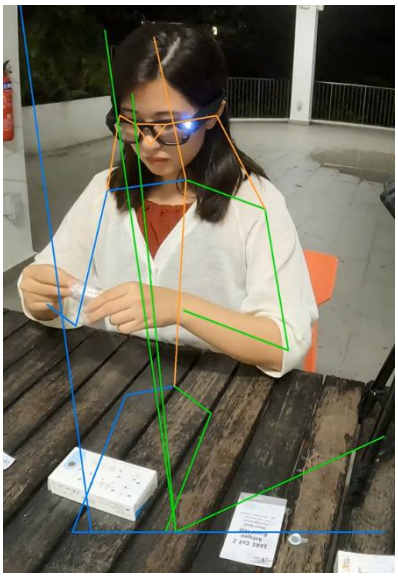
The report should include the following sections:

- **Introduction** – Describe which task you chose to tackle, providing some background in the style of a section introduction from a scientific paper.
  If the task you chose is not one of the suggested ones in this document, include a "Related Work" section, also written in the style of a scientific paper.
  If you ended up focusing on a more specific aspect of the general problem described in the assignment, add a subsection with additional details, images, and/or diagrams explaining what you focused on. Unlike a scientific paper, here you can be more descriptive and visual.
- **Methodology –** Report the main methodology you used that produced the results you are most satisfied with, explaining it clearly and as concisely as possible. Include in this section anything you implemented yourself that runs as part of your final solution. If any ideas here were taken from a paper or from AI, mention that appropriately.
- **Experiments –** Report which experiments you ran to validate your approach. The results don't have to be perfect, but they will be reflected in your grade. It is best if you can validate your approach both quantitatively (with a table of numerical values such as accuracy) and qualitatively (with exemplary results of your work). Comparing

other ideas would also improve your grade. This section should both detail how to replicate your results using your code and present the actual experimental results.

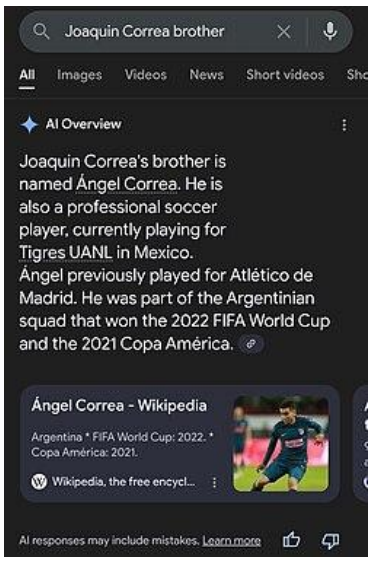
- **Work Report –** Report which parts of the code you implemented yourself. You may also describe failed attempts - if you used any of them as baselines for comparison in the previous section, explain how you implemented those attempts and what went wrong. This helps demonstrate your experimentation process and understanding, even if the results were not successful.

Project option 1 – Pose Estimation Under Occlusion.



- RTMPose is a state-of-the-art (SOTA) 2D human skeleton pose estimation family of models, trained on labeled images of humans and their joint locations.
- The currently publicly available checkpoints handle occlusions by assigning a fairly random position to the occluded limb, with very low confidence.
  An example could be seen to the right, of using the "BodyWithFeet", "performance" model from the rtmlib implementation on an image from the egoexo4d dataset.
- We would like you to create a variation of the model, in any way you choose, that predicts plausible joint locations even in these very low-confidence setups.
- A good project would explore at least 2–3 training or fine-tuning methods and provide a minimal qualitative evaluation of the improvements, as well as some results on your own images.

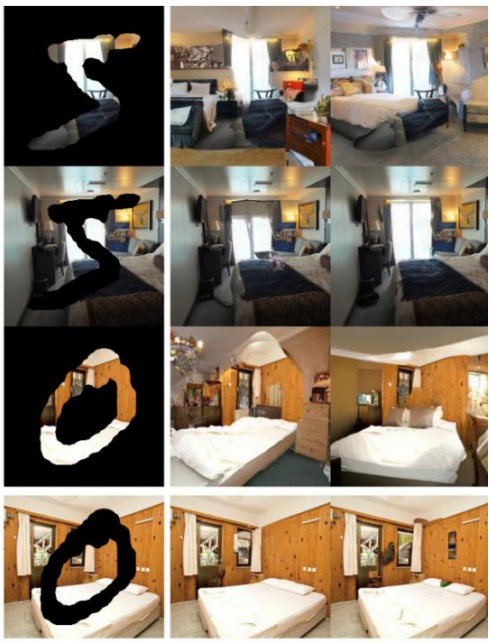
Project option 2 – Hallucination Detection.



- Large Language Models (LLMs) like ChatGPT can produce confident-sounding text with factually incorrect statements. On the right, such a statement can be found, taken from Wikipedia.
  "AI Overviews result (10 August 2025) incorrectly stating that Joaquín Correa is the brother of Ángel Correa; the two are unrelated.[28]"
- A 2025 paper and codebase published a dataset for the detection of hallucinations (and also contamination — you can ignore the DC parts).
- Use their code to generate data and evaluate their model.
- Find a way to improve their implementation, either in efficiency or in performance.
- A good project would show improvement on 2 dataset × model combinations, with no large quality loss in the other metric.

- A great project would show clear improvement with comparable results in the other metric on 4–6 dataset × model combinations.

Project option 3 – Diffusion Inpainting Implementation and Improvement.



- Diffusion inpainting is a technique that uses a DDPM-based model without additional training to fill in missing parts of an image with new content.
  The examples on the right are taken from this paper, other formulations can be found, for example, here and here by ChatGPT.
- Download stable-diffusion-2-base and implement a simple inpainting pipeline using this model. Create a few inpainting-with-text examples using your own images (not ones from the internet) and masks.
- Identify issues with the vanilla pipeline and think of ways to improve or fix these issues.
- A good project would explore an idea that improves the identified issues and demonstrate it on a few images with a minimal qualitative metric on an image dataset. A great project would explore a completely novel idea, with strong evaluations on an image dataset and multiple examples.

Alternative Projects

- You may work on your own project ideas if you wish, but it is not recommended.
- If you do so, to achieve a good grade, your project must include training a model or deploying a large model and working with a non-trivial amount of data.
- The project must demonstrate your understanding of subjects taught in the course and show an extension of the learned material in a research context.
- Since your project may cover a topic the staff is less familiar with, the report should include more details about the challenges and present related research on the topic. We will also check the related works section of the report carefully, as it will be impossible to evaluate your work without understanding it.
- With that said, we expect highly driven students who choose this path to achieve a grade similar to or higher than the rest of the class.
- You are welcome to consult with the course TA/Lecturer about your idea before you start.

Submission And "Your Work" parts.

We wish to reward students who put in effort, write their own code, and explore their own ideas. To do so, you will report what you did in the Work Report section. Furthermore, code

files submitted but not listed in the `not_our_work.txt` file will be checked for AI-generated content and will contribute positively to the final grade. If none of your work is your own, human-written code, your final grade will be lower. You are also welcome to submit any data processing scripts, as long as they are runnable and documented (either in the PDF or in the code itself).

The work must be submitted in pairs. Exemptions will be granted only in rare cases by the head TA (amir.mann@campus.technion.ac.il). Extensions will be granted for students serving in reserve duty or those with doctor-ordered hospitalization or rest, via the head TA.

As a general note, not as hard limit, if the length of your report is below 1000 words, you probably didn't do enough work or did not detail it enough. If it is above 2500 words you are probably providing too many details.

The report must be in English.